Supplementary Material for 'The carbon footprint of surgical and non-surgical skin cancer appointments'

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Supplementary Note 1. Detailed data collection methods

Data were collected between April and September 2025. Data collection forms were completed by clinic staff for a convenience sample of at least 10 dermatology appointments per clinic for each functional unit. Data collection forms were distributed using REDCap 15·3·3 (Vanderbilt University; Nashville, USA) or in printed hard copy based on staff preference. For the primary analysis, the median number of a material used was included if that material was used in more than 50% of the observed appointments.

Identified materials were sampled to record the material composition and weight of the item, its packaging, and any of its components. A digital scale (manufactured by AMIR) with a resolution of 0·1 g was used to record weight. Material composition (i.e., the type of plastic, metal, glass etc) was determined based on packaging details, public information, or assumptions based on similar materials.

The shipping of materials to Sydney, Australia was included based on an assumed shipping distance of 10,000 km and a total road-transport distance of 100 km. The sterilisation (e.g., for stainless steel surgical instruments) or washing (e.g., for patient gowns) of reusable equipment was included using data from an existing study.[1] Reusable surgical instruments were assumed to be used 300 times before being disposed and reusable patients gowns were assumed to be used 100 times before being disposed.[2,3] At the end-of-life all equipment was assumed to undergo steam autoclave sterilisation offsite, based on data from an existing study.[4] After sterilisation, equipment was assumed to be placed in landfill. Emissions from landfill were obtained from the Australian National Greenhouse Accounts Factors 2024.[5] Transport of materials between the clinic, autoclave sterilisation site, and landfill site was included. The location of sterilisation and landfill sites was obtained from the New South Wales *Protection of the Environment Operations Act* Public Register (https://apps.epa.nsw.gov.au/prpoeoapp) and distances were calculated using Google Maps (https://www.google.com.au/maps).

Patient transport was included based on data available from the MEL-SELF trial. The start location of the patient's journey to the clinic and the mode of transport was self-reported and the return travel distance calculated using Google Maps Distance Matrix API. Details of methods to calculate distance travelled by patients is available in the section 'details of methods relating to patient transport' below**.**

We included electricity for the use of a computer with internet access for the full skin examination and total body photography appointments. Emissions from heating and cooling were excluded because their consumption varies substantially by geographical location, season, and building characteristics, which would limit the generalisability of our findings.

Details of methods relating to patient transport

Patient transport was included in the model based on data available from the MELanoma SELF surveillance (MEL-SELF) randomised controlled trial, details of which are reported elsewhere.[6] Participants in the MEL-SELF trial were early-stage melanoma patients undergoing routine examination to detect changes to skin lesions. These participants are broadly representative of patients undergoing screening at the two clinics included in the current study. We retrieved data from health resource use diaries for patients enrolled in the trial at the Sydney Melanoma Diagnostic Centre (SMDC) or Melanoma Dermatology (MD) (n = 345). The health resource use diaries were monthly, retrospective, and self-reported, and contained data about the number of clinic visits made in the previous month (numeric), the origin and destination suburb of travel for those clinic visits (free text), and the mode of transport used for each visit (categorical). The destination of travel was either SMDC in Camperdown or MD in Wollstonecraft. Mode of transport was either plane, car, motorbike, bus, train, bike, or on foot.

In this analysis, we estimated the proportion of visits made using each mode of transport and the median distance travelled to attend the clinic. We only considered data from the first clinic visit reported for each patient (i.e., data for a maximum of 345

appointments). Of the 345 patients for which data were available, 262 reported details of a clinic visit at SMDC or MD over the study period.

Mode of transport

Where multiple modes of transport were reported for a single visit (e.g., a combination of car and on foot) we assumed that the entire journey was undertaken using the mode of transport with the highest environmental impact per unit of person-distance (i.e., priority was given in the following order: plane, car, motorbike, bus, train, bike, and on foot). Details are reported in **Supplementary Table 1.**

Distance travelled

Distance travelled was estimated from data about the start and end suburb of the patient's journey. The free-text start suburb was matched to an Australian post code using an open-source database.[7] For all missing data the start location was assumed to be the patient's home post code, also available from the MEL-SELF trial data. The end location was assumed to be the address of MD (40 Rocklands Road, Wollstonecraft, NSW, Australia 2065) or SMDC (Royal Prince Alfred Hospital, Missenden Road, Camperdown, NSW, Australia, 2050) as appropriate.

For journeys undertaken by car, motorbike, bus, train, bike, and on foot the distance between the start and end locations was calculated using the Google Maps Distance Matrix API and the gmapsdistance package in R.[8] This inputs for the gmapsdistance function are the start location, end location, and mode of transport. The output of the function is the distance travelled in meters. Mode of transport was inputted as either transit (for bus and train), driving (for car and motorbike), bicycling, or walking. An example snippet of code for this function is provided in **Box 1**. Distance travelled by plane was assumed to be the great circle distance between the airport closest to the patient's start post code and Sydney airport. Additional transport required for a journey by plane was not included. Details are reported in **Supplementary Table 1**.

Supplementary Note 1, Box 1. The gmapsdistance function from the gmapsdistance package in R, calculating the transit (i.e., train or bus) distance from post code 2000 (Sydney) to Melanoma Dermatology.

```
gmapsdistance(origin = "2000, Australia",
              destination = "40 Rocklands Road, Wollstonecraft, NSW,
                            Australia, 2050",
              Mode = "transit")$Distance

[1] 7678
```

References

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3 Costa D de M, Lopes LK de O, Tipple AFV, *et al.* Evaluation of stainless steel surgical instruments subjected to multiple use/processing. *Infection, Disease & Health* 2018; **23**: 3–9.

4 Zhao W, van der Voet E, Huppes G, Zhang Y. Comparative life cycle assessments of incineration and non-incineration treatments for medical waste. *Int J Life Cycle Assess* 2009; **14**: 114–21.

5 Australian Department of Climate Change, Energy, the Environment and Water. Australian National Greenhouse Accounts Factors. 2024; published online Aug. https://www.dcceew.gov.au/climate-change/publications/national-greenhouse-accounts-factors-2024.

6 Ackermann DM, Smit AK, Janda M, *et al.* Can patient-led surveillance detect subsequent new primary or recurrent melanomas and reduce the need for routinely scheduled follow-up? A protocol for the MEL-SELF randomised controlled trial. *Trials* 2021; **22**: 324.

7 Proctor M. matthewproctor/australianpostcodes. 2025; published online Sept 20. https://github.com/matthewproctor/australianpostcodes (accessed Oct 7, 2025).

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Supplementary Table 1. Distance travelled by mode of transport for a one-way journey.

	Primary analysis N=262	Sensitivity analysis[1] N=260
Primary mode of transport[2], n (%)		
Bike	1 (0.4%)	1 (0.4%)
Bus	27 (10.4%)	27 (10.5%)
Car	201 (77.6%)	201 (78.2%)
Motorbike	4 (1.5%)	4 (1.6%)
Plane	2 (0.8%)	-
Train	12 (4.6%)	12 (4.7%)
On foot	12 (4.6%)	12 (4.7%)
Unknown	3	3
Distance travelled, median (Q1, Q3) km		
Bike[3]	6 (6, 6)	6 (6, 6)
Bus[3]	10 (6, 19)	10 (6, 19)
Car[3]	15 (7, 39)	15 (7, 39)
Motorbike[3]	23 (4, 49)	23 (4, 49)
Plane[4]	408 (387, 429)	-
Train[3]	12 (6, 30)	12 (6, 30)
On foot[3]	1 (1, 1)	1 (1, 1)
Unknown[5], n	6	6

[1] Sensitivity analysis conducted excluding n=2 patients who travelled by plane

[2] Primary mode of transport used to attend first visit to SMDC or MD as reported in MEL-SELF RCT health resource use diary.

[3] Distance calculated using the Google Maps Distance Matrix API

[4] Distance calculated as the great circle distance between the closest airport to the start location and Sydney airport.

[5] Missing data occurred when the Google Maps API did not identify a route between the start and end location.

Supplementary Table 2. Life cycle assessment processes and assumptions.

Item	Weight	Process	Source	Assumptions
Alcohol swab (x1)	0·5 g	Aluminium, primary, ingot {RoW}| market for aluminium, primary, ingot | Cut-off, U	AusLCI	Item not observed directly and assumed to be the same as skin prep swab.
	0·1 g	Textile, nonwoven polypropylene {GLO}| market for textile, nonwoven polypropylene | Cut-off, U	AusLCI	
	0·5 g	Metal working, average for aluminium product manufacturing {GLO}| market for metal working, average for aluminium product manufacturing | Cut-off, U	AusLCI	
	0·6 g	Disposal, inert	Project	
	0·6 g	Shipping, to Sydney	Project	
Biopsiblade (punch) (x1)	0·1 g	Packaging film, low density polyethylene {GLO}| market for packaging film, low density polyethylene | Cut-off, U	AusLCI	
	1·7 g	Polyethylene terephthalate, granulate, amorphous {RoW}| polyethylene terephthalate, granulate, amorphous, recycled to generic market for amorphous PET granulate | Cut-off, U	AusLCI	
	2·6 g	Polypropylene, granulate {GLO}| market for polypropylene, granulate | Cut-off, U	AusLCI	
	0·1 g	Steel, chromium steel 18/8 {GLO}| market for steel, chromium steel 18/8 | Cut-off, U	AusLCI	
	1·7 g	Extrusion of plastic sheets and thermoforming, inline {GLO}| market for extrusion of plastic sheets and thermoforming, inline | Cut-off, U	AusLCI	
	2·6 g	Injection moulding {GLO}| market for injection moulding | Cut-off, U	AusLCI	
	0·1 g	Metal working, average for chromium steel product manufacturing {GLO}| market for metal working, average for chromium steel product manufacturing | Cut-off, U	AusLCI	
	4·5 g	Disposal, inert	Project	
	4·5 g	Shipping, to Sydney	Project	

Biopsiblade (shave) (x1)	0·5 g	Paper, woodfree, coated {RoW}| market for paper, woodfree, coated | Cut-off, U	AusLCI	
	2·1 g	Polyethylene terephthalate, granulate, amorphous {RoW}| polyethylene terephthalate, granulate, amorphous, recycled to generic market for amorphous PET granulate | Cut-off, U	AusLCI	
	2·2 g	Polypropylene, granulate {GLO}| market for polypropylene, granulate | Cut-off, U	AusLCI	
	0·1 g	Steel, chromium steel 18/8 {GLO}| market for steel, chromium steel 18/8 | Cut-off, U	AusLCI	
	2·1 g	Extrusion of plastic sheets and thermoforming, inline {GLO}| market for extrusion of plastic sheets and thermoforming, inline | Cut-off, U	AusLCI	
	0·1 g	Metal working, average for chromium steel product manufacturing {GLO}| market for metal working, average for chromium steel product manufacturing | Cut-off, U	AusLCI	
	0·5 g	Disposal, paper and cardboard	Project	
	4·4 g	Disposal, inert	Project	
	4·9 g	Shipping, to Sydney	Project	
Bluey	7·3 g	Polyethylene, low density, granulate {GLO}| market for polyethylene, low density, granulate | Cut-off, U	AusLCI	
	16·6 g	Cellulose fibre {CH}| market for cellulose fibre | Cut-off, U	AusLCI	
	7·3 g	Extrusion, plastic film {GLO}| market for extrusion, plastic film | Cut-off, U	AusLCI	
	16·6 g	Disposal, inert	Project	
	7·3 g	Disposal, garden and green	Project	
Cotton tipped applicator (x1)	0·5 g	Paper, woodfree, coated {RoW}| market for paper, woodfree, coated | Cut-off, U	AusLCI	Volume of sawnwood assumes density of 700kg/m^3. Weight measured was 0·9 g.
	0·5 g	Polyethylene, low density, granulate {GLO}| market for polyethylene, low density, granulate | Cut-off, U	AusLCI	

	1·29 mm^3	Sawnwood, hardwood, raw {GLO}| market for sawnwood, hardwood, raw | Cut-off, U	AusLCI	
	0·1 g	Fibre, cotton {GLO}| market for fibre, cotton | Cut-off, U	AusLCI	
	0·5 g	Extrusion, plastic film {GLO}| market for extrusion, plastic film | Cut-off, U	AusLCI	
	0·5 g	Disposal, paper and cardboard	Project	
	0·5 g	Disposal, inert	Project	
	0·9 g	Disposal, wood	Project	
	0·1 g	Disposal, textile	Project	
	2·0 g	Shipping, to Sydney	Project	
Curette (x1)	0·1 g	Packaging film, low density polyethylene {GLO}| market for packaging film, low density polyethylene | Cut-off, U	AusLCI	
	2·4 g	Polyethylene terephthalate, granulate, amorphous {RoW}| polyethylene terephthalate, granulate, amorphous, recycled to generic market for amorphous PET granulate | Cut-off, U	AusLCI	
	7·3 g	Polypropylene, granulate {GLO}| market for polypropylene, granulate | Cut-off, U	AusLCI	
	0·1 g	Steel, chromium steel 18/8 {GLO}| market for steel, chromium steel 18/8 | Cut-off, U	AusLCI	
	0·1 g	Extrusion of plastic sheets and thermoforming, inline {GLO}| market for extrusion of plastic sheets and thermoforming, inline | Cut-off, U	AusLCI	
	7·3 g	Injection moulding {GLO}| market for injection moulding | Cut-off, U	AusLCI	
	0·1 g	Metal working, average for chromium steel product manufacturing {GLO}| market for metal working, average for chromium steel product manufacturing | Cut-off, U	AusLCI	
	9·9 g	Disposal, inert	Project	
	9·9 g	Shipping, to Sydney	Project	

Disposal, garden and green (1 kg)	1 kg 1 kg	Disposal, medical waste to steam autoclave sterilisation Disposal, garden and green waste in landfill	Project Project	
Disposal, garden and green waste in landfill (1 t)	0·293 MJ 0·306 MJ 1·58E-4 kg 7·69E-3 kg 1·46E-6 kg 1·07E-2 kg 9·93E-3 kg 5·56E-7 p 1·42E-8 p 37 tkm 1·6 t 1·0E-2 kg 1·77E-2 kg 1·99E-3 kg 1·18E-4 kg	Heavy fuel oil, burned in refinery furnace {GLO}| market for heavy fuel oil, burned in refinery furnace | Cut-off, U Natural gas, burned in gas turbine {GLO}| market for natural gas, burned in gas turbine | Cut-off, U Sodium hydroxide, without water, in 50% solution state {RoW}| market for sodium hydroxide, without water, in 50% solution state | Cut-off, U Iron sulfate {RoW}| market for iron sulfate | Cut-off, U Hydrochloric acid, without water, in 30% solution state {RoW}| market for hydrochloric acid, without water, in 30% solution state | Cut-off, U Iron(II) chloride {GLO}| market for iron(II) chloride | Cut-off, U Cement, unspecified {RoW}| market for cement, unspecified | Cut-off, U Sanitary landfill facility {RoW}| sanitary landfill facility construction | Cut-off, U Wastewater treatment facility, capacity 5E9l/year {RoW}| wastewater treatment facility construction, capacity 5E9l/year | Cut-off, U Municipal waste collection service by 21 metric ton lorry {GLO}| market for municipal waste collection service by 21 metric ton lorry | Cut-off, U Carbon dioxide Nitrogen oxides, AU Sulfur dioxide, AU Ammonia, AU NMVOC, non-methane volatile organic compounds	AusLCI AusLCI AusLCI AusLCI AusLCI AusLCI AusLCI AusLCI AusLCI SP SP SP SP SP	Processes based on an existing study by Zhao et al. Transport distance calculated based on information from New South Wales Environmental Protection Agency EOAP Register. Carbon dioxide emissions based on methane data from the Australian National Greenhouse Accounts Factors.

	1·86E-3 kg	Carbon monoxide	SP	
	4·6 kg	Sulfate	SP	
	1·17 kg	Phosphate, AU	SP	
	12·7 kg	Chloride	SP	
	5·9E-2 kg	Nickel (II)	SP	
	0·56 kg	Copper, ion	SP	
	1·56 kg	Iron, ion	SP	
	1·16 kg	Zinc (II)	SP	
Disposal, inert (1 kg)	1 kg	Disposal, medical waste to steam autoclave sterilisation	Project	
	1 kg	Disposal, inert waste in landfill	Project	
Disposal, inert waste in landfill (1 t)		See 'Disposal, garden and green waste in landfill (1 t)', with only difference being carbon dioxide, as below		Carbon dioxide emissions based on methane data from the Australian National Greenhouse Accounts Factors.
	0 t	Carbon dioxide	SP	
Disposal, medical waste to steam autoclave sterilisation (430 kg)	5 kWh	Electricity, high voltage {AU-NSW}| market for electricity, high voltage | AusLCI, U	AusLCI	Processes based on an existing study by Zhao et al.
	653 kJ	Heat, district or industrial, other than natural gas {RoW}| market for heat, district or industrial, other than natural gas | Cut-off, U	AusLCI	Transport distance calculated based on information from New South Wales Environmental Protection Agency EOAP Register.
	680 kg	Tap water {RoW}| market for tap water | Cut-off, U	AusLCI	
	8·6 tkm	Municipal waste collection service by 21 metric ton lorry {GLO}| market for municipal waste collection service by 21 metric ton lorry | Cut-off, U	AusLCI	
	1·1 mg	Acetaldehyde	SP	
	0·2 mg	1-Butanol	SP	
	0·5 mg	Benzaldehyde	SP	
	7·3 mg	Ethanol	SP	
	0·2 mg	Pentane	SP	
	0·4 mg	Cyclohexanone	SP	

	0·7 mg 0·4 mg	Tetradecane D-limonene	SP SP	
Disposal, textile (1 kg)	1 kg 1 kg	Disposal, medical waste to steam autoclave sterilisation Disposal, textile waste in landfill	Project Project	
Disposal, textile waste in landfill (1 t)	 2·0 t	See 'Disposal, garden and green waste in landfill (1 t)', with only difference being carbon dioxide, as below Carbon dioxide	 SP	Carbon dioxide emissions based on methane data from the Australian National Greenhouse Accounts Factors.
Disposal, paper and cardboard (1 kg)	1 kg 1 kg	Disposal, medical waste to steam autoclave sterilisation Disposal, paper and cardboard waste in landfill	Project Project	
Disposal, paper and cardboard waste in landfill (1 t)	 3·3 t	See 'Disposal, garden and green waste in landfill (1 t)', with only difference being carbon dioxide, as below Carbon dioxide	 SP	Carbon dioxide emissions based on methane data from the Australian National Greenhouse Accounts Factors.
Disposal, wood (1 kg)	1 kg 1 kg	Disposal, medical waste to steam autoclave sterilisation Disposal, wood waste in landfill	Project Project	
Disposal, wood waste in landfill (1 t)	 0·7 t	See 'Disposal, garden and green waste in landfill (1 t)', with only difference being carbon dioxide, as below Carbon dioxide	 SP	Carbon dioxide emissions based on methane data from the Australian National Greenhouse Accounts Factors.
Drape (x1)	3·0 g 5·1 g 13·4 g 8·4 g	Packaging film, low density polyethylene {GLO}| market for packaging film, low density polyethylene | Cut-off, U Paper, woodfree, coated {RoW}| market for paper, woodfree, coated | Cut-off, U Textile, nonwoven polypropylene {GLO}| market for textile, nonwoven polypropylene | Cut-off, U Polyethylene, low density, granulate {GLO}| market for polyethylene, low density, granulate | Cut-off, U	AusLCI AusLCI AusLCI AusLCI	Textile and polyethylene layers of the drape could not be separated. Total weight (16·8 g) was halved for each layer (8·4 g).

	8·4 g	Extrusion, plastic film {GLO}| market for extrusion, plastic film | Cut-off, U	AusLCI	
	24·8 g	Disposal, inert	Project	
	5·1 g	Disposal, paper and cardboard	Project	
	29·9 g	Shipping, to Sydney	Project	
Dressing (x1)	0·5 g	Packaging film, low density polyethylene {GLO}| market for packaging film, low density polyethylene | Cut-off, U	AusLCI	
	0·5 g	Paper, woodfree, coated {RoW}| market for paper, woodfree, coated | Cut-off, U	AusLCI	
	0·4 g	Textile, nonwoven polypropylene {GLO}| market for textile, nonwoven polypropylene | Cut-off, U	AusLCI	
	0·2 g	Polyethylene, low density, granulate {GLO}| market for polyethylene, low density, granulate | Cut-off, U	AusLCI	
	0·2 g	Extrusion, plastic film {GLO}| market for extrusion, plastic film | Cut-off, U	AusLCI	
	1·1 g	Disposal, inert	Project	
	0·5 g	Disposal, paper and cardboard	Project	
	1·6 g	Shipping, to Sydney	Project	
Dressing Pack (x1)	1·7 g	Paper, woodfree, coated {RoW}| market for paper, woodfree, coated | Cut-off, U	AusLCI	Textile and polyethylene layers of the inner packaging and antiseptic field could not be separated. Total weight (9·1 g and 6·5 g, respectively) was halved for each layer (4·55 g and 3·25 g, respectively).
	1·6 g	Packaging film, low density polyethylene {GLO}| market for packaging film, low density polyethylene | Cut-off, U	AusLCI	
	11·9 g	Textile, nonwoven polypropylene {GLO}| market for textile, nonwoven polypropylene | Cut-off, U	AusLCI	
	7·8 g	Polyethylene, low density, granulate {GLO}| market for polyethylene, low density, granulate | Cut-off, U	AusLCI	
	8·7 g	Polystyrene, general purpose {GLO}| market for polystyrene, general purpose | Cut-off, U	AusLCI	
	10.5 g	Polypropylene, granulate {GLO}| market for polypropylene, granulate | Cut-off, U	AusLCI	

	6·5 g	Extrusion, plastic film {GLO}| market for extrusion, plastic film | Cut-off, U	AusLCI	
	8·7 g	Extrusion of plastic sheets and thermoforming, inline {GLO}| market for extrusion of plastic sheets and thermoforming, inline | Cut-off, U	AusLCI	
	10·5 g	Injection moulding {GLO}| market for injection moulding | Cut-off, U	AusLCI	
	1·7 g	Disposal, paper and cardboard	Project	
	40·5 g	Disposal, inert	Project	
	42·2 g	Shipping, to Sydney	Project	
Electricity, with computers (1 min)	1 min	Operation, computer, desktop, with liquid crystal display, active mode {GLO}| market for operation, computer, desktop, with liquid crystal display, active mode | Cut-off, U	AusLCI	
	1 min	Internet access, work, 0.2 Mbit/s {GLO}| market for internet access, work, 0.2 Mbit/s | Cut-off, U	AusLCI	
Examination bed sheet (x1)	50·1 g	Tissue paper {GLO}| market for tissue paper | Cut-off, U	AusLCI	
	50·1 g	Disposal, paper and cardboard	Project	
	50·1 g	Shipping, to Sydney	Project	
Forceps (x1)	2·5 g	Polyethylene, low density, granulate {GLO}| market for polyethylene, low density, granulate | Cut-off, U	AusLCI	Instrument assumed to be used 300 times over lifetime.
	3·5 g	Paper, woodfree, uncoated {RoW}| market for paper, woodfree, uncoated | Cut-off, U	AusLCI	Packaging material was assumed to be the same as Gillies Skin Hook.
	0·221 g	Steel, chromium steel 18/8 {GLO}| market for steel, chromium steel 18/8 | Cut-off, U	AusLCI	
	2·5 g	Extrusion, plastic film {GLO}| market for extrusion, plastic film | Cut-off, U	AusLCI	
	0·221 g	Metal working, average for chromium steel product manufacturing {GLO}| market for metal working, average for chromium steel product manufacturing | Cut-off, U	AusLCI	
	29·0 g	Sterilisation of reusable instrument	Project	

	2·721 g	Disposal, inert	Project	
	3·5 g	Disposal, paper and cardboard	Project	
	6·221 g	Shipping, to Sydney	Project	
Gallipot (x1)	2·4 g	Packaging film, low density polyethylene {GLO}| market for packaging film, low density polyethylene | Cut-off, U	AusLCI	
	1·2 g	Paper, woodfree, coated {RoW}| market for paper, woodfree, coated | Cut-off, U	AusLCI	
	10·0 g	Polypropylene, granulate {GLO}| market for polypropylene, granulate | Cut-off, U	AusLCI	
	10·0 g	Injection moulding {GLO}| market for injection moulding | Cut-off, U	AusLCI	
	12·4 g	Disposal, inert	Project	
	1·2 g	Disposal, paper and cardboard	Project	
	13·6 g	Shipping, to Sydney	Project	
Gauze (x5)	1·4 g	Packaging film, low density polyethylene {GLO}| market for packaging film, low density polyethylene | Cut-off, U	AusLCI	
	3·7 g	Textile, woven cotton {GLO}| market for textile, woven cotton | Cut-off, U	AusLCI	
	1·4 g	Disposal, inert	Project	
	3·7 g	Disposal, textile	Project	
	5·1 g	Shipping, to Sydney	Project	
Gillies skin hook (1x)	2·5 g	Polyethylene, low density, granulate {GLO}| market for polyethylene, low density, granulate | Cut-off, U	AusLCI	Instrument assumed to be used 300 times over lifetime
	3·5 g	Paper, woodfree, uncoated {RoW}| market for paper, woodfree, uncoated | Cut-off, U	AusLCI	
	0·204 g	Steel, chromium steel 18/8 {GLO}| market for steel, chromium steel 18/8 | Cut-off, U	AusLCI	
	2·5 g	Extrusion, plastic film {GLO}| market for extrusion, plastic film | Cut-off, U	AusLCI	

	0·204 g	Metal working, average for chromium steel product manufacturing {GLO}| market for metal working, average for chromium steel product manufacturing | Cut-off, U	AusLCI	
	20·4 g	Sterilisation of reusable instrument	Project	
	2·704 g	Disposal, inert	Project	
	3·5 g	Disposal, paper and cardboard	Project	
	6·204 g	Shipping, to Sydney	Project	
Kidney tray (x1)	26·6 g	Bagasse, from sugarcane {RoW}| market for bagasse, from sugarcane | Cut-off, U	AusLCI	
	26·6 g	Disposal, garden and green	Project	
	26·6 g	Shipping, to Sydney	Project	
Laundry (1 t)	192·69 kWh	Electricity, low voltage {AU-NSW}| market for electricity, low voltage | AusLCI, U	AusLCI	Private correspondence, Linen Services Australia
	4·33 GJ	Heat, central or small-scale, natural gas {RoW}| market for heat, central or small-scale, natural gas | Cut-off, U	AusLCI	
	7350 kg	Tap water {RoW}| market for tap water | Cut-off, U	AusLCI	
	1·5 kg	Ethoxylated alcohol (AE>20) {GLO}| market for ethoxylated alcohol (AE>20) | Cut-off, U	AusLCI	
Needle (x1)	0·1 g	Packaging film, low density polyethylene {GLO}| market for packaging film, low density polyethylene | Cut-off, U	AusLCI	
	0·1 g	Paper, woodfree, coated {RoW}| market for paper, woodfree, coated | Cut-off, U	AusLCI	
	0·8 g	Polypropylene, granulate {GLO}| market for polypropylene, granulate | Cut-off, U	AusLCI	
	0·1 g	Steel, chromium steel 18/8 {GLO}| market for steel, chromium steel 18/8 | Cut-off, U	AusLCI	
	0·8 g	Injection moulding {GLO}| market for injection moulding | Cut-off, U	AusLCI	

	0·1 g	Metal working, average for chromium steel product manufacturing {GLO}| market for metal working, average for chromium steel product manufacturing | Cut-off, U	AusLCI	
	1·0 g	Disposal, inert	Project	
	0·1 g	Disposal, paper and cardboard	Project	
	1·1 g	Shipping, to Sydney	Project	
Needle holder (x1)	2·5 g	Polyethylene, low density, granulate {GLO}| market for polyethylene, low density, granulate | Cut-off, U	AusLCI	Instrument assumed to be used 300 times over lifetime.
	3·5 g	Paper, woodfree, uncoated {RoW}| market for paper, woodfree, uncoated | Cut-off, U	AusLCI	Packaging material was assumed to be the same as Gillies Skin Hook.
	0·294 g	Steel, chromium steel 18/8 {GLO}| market for steel, chromium steel 18/8 | Cut-off, U	AusLCI	
	2·5 g	Extrusion, plastic film {GLO}| market for extrusion, plastic film | Cut-off, U	AusLCI	
	0·294 g	Metal working, average for chromium steel product manufacturing {GLO}| market for metal working, average for chromium steel product manufacturing | Cut-off, U	AusLCI	
	29·4 g	Sterilisation of reusable instrument	Project	
	2·794 g	Disposal, inert	Project	
	3·5 g	Disposal, paper and cardboard	Project	
	6·294 g	Shipping, to Sydney	Project	
Nonsterile glove (x1)	2·7 g	Acrylonitrile {GLO}| market for acrylonitrile | Cut-off, U	AusLCI	
	2·7 g	Disposal, inert	Project	
	2·7 g	Shipping, to Sydney	Project	
Paper (x1)	5·1 g	Paper, woodfree, uncoated {RoW}| market for paper, woodfree, uncoated | Cut-off, U	AusLCI	
	5·1 g	Disposal, paper and cardboard	Project	
	5·1 g	Shipping, to Sydney	Project	

Paper towel (x1)	2·0 g	Tissue paper {GLO}| market for tissue paper | Cut-off, U	AusLCI	
	2·0 g	Disposal, paper and cardboard	Project	
	2·0 g	Shipping, to Sydney	Project	
Patient gown (x1)	330·1 g	Fibre, cotton {GLO}| market for fibre, cotton | Cut-off, U	AusLCI	Gown assumed to be used 100 times over lifetime.
	330·1 g	Laundry	Project	
	3·3 g	Disposal, textile	Project	
	3·3 g	Shipping, to Sydney	Project	
Plastic bag (x1)	9·6 g	Polyethylene, high density, granulate {RoW}| polyethylene, high density, granulate, recycled to generic market for high density PE granulate | Cut-off, U	AusLCI	
	9·6 g	Extrusion, plastic film {GLO}| market for extrusion, plastic film | Cut-off, U	AusLCI	
	9·6 g	Disposal, inert	Project	
	9·6 g	Shipping, to Sydney	Project	
Razor (x1)	5·0 g	Polystyrene, general purpose {GLO}| market for polystyrene, general purpose | Cut-off, U	AusLCI	
	0·1 g	Steel, chromium steel 18/8 {GLO}| market for steel, chromium steel 18/8 | Cut-off, U	AusLCI	
	5·0 g	Injection moulding {GLO}| market for injection moulding | Cut-off, U	AusLCI	
	0·1 g	Metal working, average for chromium steel product manufacturing {GLO}| market for metal working, average for chromium steel product manufacturing | Cut-off, U	AusLCI	
	5·1 g	Disposal, inert	Project	
	5·1 g	Shipping, to Sydney	Project	
Scalpel handle (x1)	2·5 g	Polyethylene, low density, granulate {GLO}| market for polyethylene, low density, granulate | Cut-off, U	AusLCI	Instrument assumed to be used 300 times over lifetime.

	3·5 g	Paper, woodfree, uncoated {RoW}| market for paper, woodfree, uncoated | Cut-off, U	AusLCI	Packaging material was assumed to be the same as Gillies Skin Hook.
	0·258 g	Steel, chromium steel 18/8 {GLO}| market for steel, chromium steel 18/8 | Cut-off, U	AusLCI	
	2·5 g	Extrusion, plastic film {GLO}| market for extrusion, plastic film | Cut-off, U	AusLCI	
	0·258 g	Metal working, average for chromium steel product manufacturing {GLO}| market for metal working, average for chromium steel product manufacturing | Cut-off, U	AusLCI	
	25·8 g	Sterilisation of reusable instrument	Project	
	2·758 g	Disposal, inert	Project	
	3·5 g	Disposal, paper and cardboard	Project	
	6·258 g	Shipping, to Sydney	Project	
Scissors (x1)	2·5 g	Polyethylene, low density, granulate {GLO}| market for polyethylene, low density, granulate | Cut-off, U	AusLCI	Instrument assumed to be used 300 times over lifetime.
	3·5 g	Paper, woodfree, uncoated {RoW}| market for paper, woodfree, uncoated | Cut-off, U	AusLCI	Packaging material was assumed to be the same as Gillies Skin Hook.
	0·290 g	Steel, chromium steel 18/8 {GLO}| market for steel, chromium steel 18/8 | Cut-off, U	AusLCI	
	2·5 g	Extrusion, plastic film {GLO}| market for extrusion, plastic film | Cut-off, U	AusLCI	
	0·290 g	Metal working, average for chromium steel product manufacturing {GLO}| market for metal working, average for chromium steel product manufacturing | Cut-off, U	AusLCI	
	29·0 g	Sterilisation of reusable instrument	Project	
	2·790 g	Disposal, inert	Project	
	3·5 g	Disposal, paper and cardboard	Project	
	6·290 g	Shipping, to Sydney	Project	
Shipping, to Sydney (1 kg)	10,000 kgkm	Transport, freight, sea, container ship {GLO}| market for transport, freight, sea, container ship | Cut-off, U	AusLCI	Assumes transport distance of 10,000 km

	100 kgkm	Transport, freight, lorry, unspecified {RoW}| market for transport, freight, lorry, unspecified | Cut-off, U	AusLCI	by ship and 100 km by road
Skin prep swab (x1)	0·5 g	Aluminium, primary, ingot {RoW}| market for aluminium, primary, ingot | Cut-off, U	AusLCI	
	0·1 g	Textile, nonwoven polypropylene {GLO}| market for textile, nonwoven polypropylene | Cut-off, U	AusLCI	
	0·5 g	Metal working, average for aluminium product manufacturing {GLO}| market for metal working, average for aluminium product manufacturing | Cut-off, U	AusLCI	
	0·6 g	Disposal, inert	Project	
	0·6 g	Shipping, to Sydney	Project	
Specimen bag (x1)	5·3 g	Polyethylene, low density, granulate {GLO}| market for polyethylene, low density, granulate | Cut-off, U	AusLCI	
	5·3 g	Extrusion, plastic film {GLO}| market for extrusion, plastic film | Cut-off, U	AusLCI	
	5·3 g	Disposal, inert	Project	
	5·3 g	Shipping, to Sydney	Project	
Specimen jar (x1)	12·6 g	Polypropylene, granulate {GLO}| market for polypropylene, granulate | Cut-off, U	AusLCI	
	12·6 g	Injection moulding {GLO}| market for injection moulding | Cut-off, U	AusLCI	
	12·6 g	Disposal, inert	Project	
	12·6 g	Shipping, to Sydney	Project	
Sterilisation of reusable instrument (1 kg)	0·9 kWh	Electricity, low voltage {AU-NSW}| market for electricity, low voltage | AusLCI, U	AusLCA	Processes based on an existing study by McGain et al.
Surgical blade (x1)	0·5 g	Aluminium, primary, ingot {RoW}| market for aluminium, primary, ingot | Cut-off, U	AusLCI	
	0·4 g	Steel, low-alloyed {GLO}| market for steel, low-alloyed | Cut-off, U	AusLCI	

	0·5 g	Metal working, average for aluminium product manufacturing {GLO}| market for metal working, average for aluminium product	AusLCI	
	0·4 g	Metal working, average for chromium steel product manufacturing {GLO}| market for metal working, average for chromium steel product manufacturing | Cut-off, U	AusLCI	
	0·9 g	Disposal, inert	Project	
	0·9 g	Shipping, to Sydney	Project	
Surface wipe	1·1 g	Textile, nonwoven polypropylene {GLO}| market for textile, nonwoven polypropylene | Cut-off, U	AusLCI	
	1·1 g	Disposal, inert	Project	
	1·1 g	Shipping, to Australia	Project	
Surgical mask (x1)	4·0 g	Textile, nonwoven polypropylene {GLO}| market for textile, nonwoven polypropylene | Cut-off, U	AusLCI	
	4·0 g	Disposal, inert	Project	
	4·0 g	Shipping, to Sydney	Project	
Suture (x1)	2·5 g	Aluminium, primary, ingot {RoW}| market for aluminium, primary, ingot | Cut-off, U	AusLCI	
	2·4 g	Paper, woodfree, coated {RoW}| market for paper, woodfree, coated | Cut-off, U	AusLCI	
	0·1 g	Nylon 6 {RoW}| market for nylon 6 | Cut-off, U	AusLCI	
	0·1 g	Steel, chromium steel 18/8 {GLO}| market for steel, chromium steel 18/8 | Cut-off, U	AusLCI	
	2·5 g	Metal working, average for aluminium product manufacturing {GLO}| market for metal working, average for aluminium product manufacturing | Cut-off, U	AusLCI	
	0·1 g	Extrusion, plastic pipes {GLO}| market for extrusion, plastic pipes | Cut-off, U	AusLCI	
	0·1 g	Metal working, average for chromium steel product manufacturing {GLO}|	AusLCI	

		market for metal working, average for chromium steel product manufacturing | Cut-off, U		
	2·7 g	Disposal, inert	Project	
	2·4 g	Disposal, paper and cardboard	Project	
	5·1 g	Shipping, to Sydney	Project	
Sterile gloves (x2)	1·4 g	Polyethylene, high density, granulate {RoW}| polyethylene, high density, granulate, recycled to generic market for high density PE granulate | Cut-off, U	AusLCI	
	1·4 g	Packaging film, low density polyethylene {GLO}| market for packaging film, low density polyethylene | Cut-off, U	AusLCI	
	4·3 g	Paper, woodfree, coated {RoW}| market for paper, woodfree, coated | Cut-off, U	AusLCI	
	21·9 g	Acrylonitrile {GLO}| market for acrylonitrile | Cut-off, U	AusLCI	
	1·4 g	Extrusion, plastic film {GLO}| market for extrusion, plastic film | Cut-off, U	AusLCI	
	24·7 g	Disposal, inert	Project	
	4·3 g	Disposal, paper and cardboard	Project	
	29.0 g	Shipping, to Sydney	Project	
Steristrip (x1)	1·0 g	Packaging film, low density polyethylene {GLO}| market for packaging film, low density polyethylene | Cut-off, U	AusLCI	
	0·2 g	Polyethylene, low density, granulate {GLO}| market for polyethylene, low density, granulate | Cut-off, U	AusLCI	
	0·1 g	Textile, nonwoven polypropylene {GLO}| market for textile, nonwoven polypropylene | Cut-off, U	AusLCI	
	0·2 g	Extrusion, plastic film {GLO}| market for extrusion, plastic film | Cut-off, U	AusLCI	
	1·3 g	Disposal, inert	Project	
	1·3 g	Shipping, to Sydney	Project	

Syringe (x1)	0·9 g	Packaging film, low density polyethylene {GLO}| market for packaging film, low density polyethylene | Cut-off, U	AusLCI	
	3·6 g	Polypropylene, granulate {GLO}| market for polypropylene, granulate | Cut-off, U	AusLCI	
	0·1 g	Synthetic rubber {GLO}| market for synthetic rubber | Cut-off, U	AusLCI	
	3·7 g	Injection moulding {GLO}| market for injection moulding | Cut-off, U	AusLCI	
	4·5 g	Disposal, inert	Project	
	0·1 g	Disposal, rubber	Project	
	4·6 g	Shipping, to Sydney	Project	
Tape (1 cm)	1·8 g	Textile, nonwoven polypropylene {GLO}| market for textile, nonwoven polypropylene | Cut-off, U	AusLCI	
	1.8 g	Disposal, inert	Project	
	1.8 g	Shipping, to Sydney	Project	
Tissue (x1)	1·2 g	Tissue paper {GLO}| market for tissue paper | Cut-off, U	AusLCI	
	1·2 g	Disposal, paper and cardboard	Project	
	1·2 g	Shipping, to Sydney	Project	
Tray (x1)	0·78 g	Polypropylene, granulate {GLO}| market for polypropylene, granulate | Cut-off, U	AusLCI	
	0·14 g	Steel, chromium steel 18/8 {GLO}| market for steel, chromium steel 18/8 | Cut-off, U	AusLCI	
	0·78 g	Injection moulding {GLO}| market for injection moulding | Cut-off, U	AusLCI	
	0·14 g	Metal working, average for chromium steel product manufacturing {GLO}| market for metal working, average for chromium steel product manufacturing | Cut-off, U	AusLCI	
	91·97 g	Sterilisation of reusable instrument	Project	
	0·92 g	Disposal, inert	Project	
	0·92 g	Shipping, to Sydney	Project	

SP: SimaPro

Supplementary Table 3. Data collection completion for each functional unit.

	FSE, n=20	TBP, n=20	Excision, n=26
Clinic, n (%)			
SMDC	10 (50%)	10 (50%)	8 (31%)
MD	10 (50%)	10 (50%)	18 (69%)

MD: Melanoma Dermatology, FSE: Full skin examination, SMDC: Sydney Melanoma Diagnostic Centre, TBP: total body photography

Supplementary Table 4. Length of clinic appointment (mins) for each functional unit.

	FSE, n=20	TBP, n=20	Excision, n=26
Appointment duration, median (min, max) minutes	30.5 (5, 50)	20 (20, 40)	30 (12, 51)

FSE: Full skin examination, TBP: total body photography

Supplementary Table 5. Reusable and disposable materials used during an appointment for a full skin examination. Materials shaded green were used in more than 50% of appointments and were included in the primary analysis. All materials were included in the Monte Caro simulation.

	Full skin examination, n=20	
	Median (min, max)	**N (%)[1]**
Single-use equipment,		
Examination bed sheet	1 (1, 1)	20 (100%)
Alcohol wipe	1 (1, 1)	20 (100%)
Paper towel	0 (0, 2)	9 (45%)
Printer paper	0 (0, 2)	6 (30%)
Surface wipe	0 (0, 2)	3 (15%)
Tissue	0 (0, 1)	1 (5%)
Glove, non-sterile	0 (0, 2)	2 (10%)

MD: Melanoma Dermatology, SMDC: Sydney Melanoma Diagnostic Centre

[1] Number (%) of appointments where item was used

Supplementary Table 6. Reusable and disposable materials used during an appointment for total body photography. Materials shaded green were used in more than 50% of appointments and were included in the primary analysis. All materials were included in the Monte Caro simulation.

	Total body photography, n=20	
	Median (min, max)	**N (%)[1]**
Reusable equipment		
Patient gown	0 (0, 1)	8 (40%)
Single-use equipment		
Paper towel	0.5 (0, 1)	10 (50%)
Surface wipe	0 (0, 1)	4 (20%)

MD: Melanoma Dermatology, SMDC: Sydney Melanoma Diagnostic Centre

[1] Number (%) of appointments where item was used

Supplementary Table 7. Reusable and disposable materials used during an appointment for the excision of a lesion suspicious for melanoma or other skin cancer. Data includes all types and concurrent number of excisions. Materials shaded green were used in more than 50% of appointments and were included in the primary analysis. All materials were included in the Monte Caro simulation.

	Excision of lesion, n=26	
	Median (min, max)	**N (%)[1]**
Reusable equipment		
Needle holder	1 (0, 1)	22 (85%)
Forceps	1 (0, 2)	22 (85%)
Scissor	2 (0, 2)	20 (77%)
Scalpel handle	1 (0, 1)	15 (58%)
Tray	0 (0, 1)	8 (31%)
Gillies skin hook	0 (0, 1)	1 (4%)
Single-use equipment		
Skin prep swab	1 (1, 1)	26 (100%)
Specimen jar	1·5 (1, 5)	26 (100%)
Gauze	15 (5, 30)	26 (100%)
Syringe	1 (1, 10)	26 (100%)
Specimen bag	2 (0, 2)	25 (96%)
Nonsterile glove	2 (0, 8)	25 (96%)
Kidney tray	1 (0, 1)	24 (92%)
Tape	15 (0, 50)	23 (88%)
Suture	1·5 (0, 4)	22 (85%)
Dresing	1 (0, 5)	21 (81%)
Surgical mask	2 (0, 2)	20 (77%)
Sterile glove	2 (0, 4)	18 (69%)
Dressing pack[2]	1 (0, 1)	18 (69%)
Paper	1 (0, 2)	18 (69%)
Drape	1 (0, 2)	17 (65%)
Tissue	6 (0, 8)	17 (65%)
Biopsiblade (punch)	1 (0, 2)	15 (58%)
Blueys	1 (0, 2)	15 (58%)
Surgical blade	0 (0, 1)	10 (38%)

Needle	0 (0, 2)	9 (35%)
Surface wipe	0 (0, 1)	8 (31%)
Gallipot	0 (0, 1)	8 (31%)
Cotton tipped applicator	0 (0, 4)	7 (27%)
Biopsiblade (shave)	0 (0, 1)	6 (23%)
Examination bed sheet	0 (0, 1)	6 (23%)
Razor	0 (0, 1)	5 (19%)
Steristrip	0 (0, 1)	4 (15%)
Plastic bag	0 (0, 2)	2 (8%)
Curette	0 (0, 1)	1 (4%)

MD: Melanoma Dermatology, SMDC: Sydney Melanoma Diagnostic Centre

[1] Number (%) of appointments where item was used

[2] Contains disposable antiseptic field, tray, forceps, and gauze.

Supplementary Table 8. Number and type of excision performed during n=26 appointments for the excision of a lesion suspicious for melanoma or other skin cancer. Data includes all types and concurrent number of excisions.

	Excision of lesion, n=26
Number of lesions removed, n (%)	
1	13 (50%)
2	8 (31%)
3	2 (7·7%)
4	2 (7·7%)
5	1 (3·8%)
Type of excision, n (%)[1]	
Excision	17 (35%)
Shave	7 (15%)
Punch	24 (50%)

[1] Percentage of all excisions (n=48) not number of appointments (n=26).

Supplementary Table 9. Details of material composition and weight for single-use and reusable instruments.

Name	Detail	Component	Material	Weight
		Single-use instruments		
Syringe	Terumo syringe 5cc/mL LUER Lock Tip	Packaging	LDPE	0·9 g
		Plunger	PPE	1·5 g
		Barrell	PPE	2·1 g
		Seal	Synthetic rubber	<0·1 g[1]
Gauze	Sterile gauze swab white 7·5cm x 7·5cm 8 ply, 5 pack	Packaging	LDPE	1·4 g
		Gauze (x5)	Cotton	3·7 g
Drape	Drape with oval aperture 45cm x 60cm, aperture 8cm x 6cm	Packaging	LDPE	3·0 g
		Packaging	Paper	2·7 g
		Insert	Paper	2·4 g
		Drape	PPE textile and LDPE	16·8 g
		Drape packaging	PPE textile	5·0 g
Dressing	Dressing, non-woven island, 7·5x5cm	Packaging	LDPE	0·5 g
		Packaging	Paper	0·5 g
		Dressing	PPE textile	0·4 g
		Dressing back	LDPE	0·2 g
Specimen jar	-	Jar	PPE	7·8 g
		Lid	PPE	4·8 g
Curette	Dermal curette 4mm	Packaging	LDPE	<0·1 g[1]
		Packaging	PET	2·4 g
		Curette	PPE	7·3 g
		Blade	Steel	<0·1 g[1]
Gallipot	Gallipot, 10mL	Packaging	LDPE	2·4 g
		Packaging	Paper	1·2 g
		Gallipot	PPE	10·0 g
Nonsterile glove	Non-sterile nitrile glove, XS	Glove	Acrylonitrile	2·7 g
Sterile glove	Gammex non-latex sensitive, size 6·5	Packaging	HDPE	1·4 g
		Packaging	LDPE	1·4 g
		Insert	Paper	4·3 g
		Gloves (x2)	Acrylonitrile	21·9 g
Surgical mask	-	Mask	PPE textile	4·0 g
Specimen bag	-	Bag	LDPE	5·3 g
Needle	Terumo Agani Needle 18G LUER 6% Bevel	Packaging	LDPE	0·1 g
		Packaging	Paper	0·1 g
		Sheath	PPE	0·7 g

		Needle base	PPE	<0·1 g[1]
		Needle cannula	Steel	<0·1 g[1]
Cotton tipped applicator	Cotton tipped applicator, 15cm, 2pcs	Packaging	Paper	0·5 g
		Packaging	LDPE	0·5 g
		Applicator handle	Sawnwood	0·9 g
		Tip	Cotton	<0·1 g[1]
Razor	Bic body razor	Razor	PS	4·2 g
		Cover	PS	0·8 g
		Blade	Steel	0·1 g[2]
Biopsiblade (shave)	-	Packaging	Paper	0·5 g
		Packaging	PET	2·1 g
		Biopsiplade	PPE	2·2 g
		Blade	Steel	0·1 g[2]
Biopsiblade (punch)	Disposable biopsy punch	Packaging	LDPE	<0·1 g[1]
		Packaging	PET	1·7 g
		Biopsy punch	PPE	2·6 g
		Blade	Steel	0·1 g[2]
Surgical blade	Surgical blade, No 15 carbon steel	Packaging	Aluminium	0·5 g
		Blade	Steel	0·4 g
Steristrip	-	Packaging	LDPE	1·0 g
		Backing	LDPE	0·2 g
		Steri-strip	PPE textile	<0·1 g[1]
Skin prep swab	Chlorohexidine skin prep swap	Packaging	Aluminium	0·5 g
		Wipe[3]	PPE textile	<0·1 g[1]
Alcohol wipe[4]	-	-	-	-
Tape	Micropore tape, 1m	Tape	PPE textile	1·8 g
Kidney tray	-	Tray	Bagasse	26·6 g
Dressing pack	Multigate SmartTab Basic Dressing Pack	Packaging	Paper	1·7 g
		Packaging	LDPE	1·6 g
		Packaging	PPE textile and LDPE	9·1 g
		Antiseptic field	PPE textile and LDPE	3·25 g
		Tray	PS	8·7 g
		Gauze	PPE textile	4·1 g
		Forceps x3	PPE	10·5 g
Suture	Suture PD synth 3-0 45cm	Packaging	Aluminium	2·5 g
		Insert	Paper	2·4 g
		Suture	Nylon	<0·1 g[1]
		Cutting needle	Steel	<0·1 g[1]
Tissue	-	Tissue	Paper	1·2 g
Printer paper	A4 printer paper	Paper	Paper	5·1 g
Examination bed sheet	Examination bed sheet, 180 cm	Paper	Paper	50·1 g

Paper towel	-	Paper	Paper	2·0 g
Surface wipe	-	Wipe (dry)	PPE textile	1·1 g
Bluey	-	Plastic layer	LDPE	7·3 g
		Textile layer	Cellulose	16·6 g
Plastic bag	-	Bag	HDPE	9·6 g
Reusable instruments				
Gillies skin hook	-	Packaging[5]	LDPE	2·5 g
		Packaging[5]	Paper	2·1 g
		Skin hook	Steel	20·4 g
		Insert[5]	Paper	1·4 g
Needle holder[6]	-	Needle holder	Steel	29·4 g[7]
Scissor[6]	-	Scissors	Steel	29·0 g[8]
Forceps[6]	-	Forceps	Steel	22·1 g[9]
Scalpel handle[6]	-	Scalpel handle	Steel	25·8 g[10]
Patient gown	-	Gown	Cotton	330·1 g
Tray	-	Tray	PPE	77·97 g[11]
		Tag	Steel	14·0 g[12]

LDPE: low-density polyethylene, PET: polyethylene terephthalate, PPE: polypropylene, PS: polystyrene

[1] Components that weighed less that 0·1 g were included in the model with a weight of 0·1 g
[2] Blade could not be isolated and measured directly and was include in the model as 0·1 g
[3] Wipe weighed 0·5 g when wet and <0·1 g when dry. Dry weight was used.
[4] Material was not measured directly and assumed to be the same as the skin prep swab.
[5] Component is not reused.
[6] Packaging was not measured directly. Estimates are based on the mean weight of the appropriate reusable instrument as measured in examined excision sets (which contain multiple instruments) and the packaging and insert weight of the Gillies skin hook (which was examined directly).
[7] Mean of needle holder weights from three examined instrument kits: 28·3 g, 26·8 g, 33·0 g
[8] Mean of scissors weights from two examined kits: 23·8 g, 34·1 g
[9] Mean of forceps weights from three examined kits: 22·9 g, 21·2 g, 22·1 g
[10] Mean of scalpel handle weights from two examined kits: 23·7 g, 27·9 g
[11] Mean of tray weights from three examined kits: 98·1g, 94·0 g, 41·8 g
[12] Mean of tag weights from three examined kits: 13·5 g, 14·4 g

Supplementary Table 10. Reusable and disposable materials used during an appointment for the excision of one lesion suspicious for melanoma or other skin cancer. Data excludes all instances where more than one lesion was removed during the appointment. Data is reported by the type of procedure (excision, punch, or shave). Materials shaded green were used in more than 50% of appointments and were included in the primary analysis. All materials were included in the Monte Caro simulation.

	1x Excision (n=3)	**1x Punch (n=7)**	**1x Shave (n=3)**
Clinic, n (%)			
SMDC	1 (33%)	5 (71%)	0 (0%)
MD	2 (67%)	2 (29%)	3 (100%)
Appointment duration, minutes			
Median (min, max)	30 (20, 50)	16 (12, 30)	20 (15, 20)
Reusable equipment, median (min, max) number used per appointment; n (%) appointments where item was used			
Needle holder	1 (1, 1); 2 (67%)	1 (1, 1); 7 (100%)	-
Scissor	2 (2, 2); 3 (100%)	2 (0, 2); 5 (71%)	-
Forceps	1 (1, 2); 3 (100%)	1 (1, 2); 7 (100%)	-
Scalpel handle	1 (1, 1); 3 (100%)	0 (0, 1); 3 (43%)	-
Tray	0 (0,1); 1 (33%)	1 (0, 1); 5 (71%)	-
Single-use equipment, median (min, max) number per appointment; n (%) appointments where item was used			
Skin prep swab	1 (1, 1); 3 (100%)	1 (1, 1); 7 (100%)	1 (1, 1); 3 (100%)
Drape	1 (1, 1); 3 (100%)	0 (0, 1); 3 (43%)	-
Scalpel blade	1 (1, 1); 3 (100%)	-	-
Specimen jar	1 (1, 1); 3 (100%)	1 (1, 1); 7 (100%)	1 (1, 1); 3 (100%)

Specimen bag	2 (1, 2); 3 (100%)	1 (0, 2); 6 (86%)	2 (2, 2); 3 (100%)
Suture	2 (2, 2); 3 (100%)	1 (1, 2); 7 (100%)	-
Surgical mask	2 (2, 2); 3 (100%)	1 (0, 2); 6 (86%)	-
Sterile gloves	2 (2, 2); 3 (100%)	2 (0, 2); 5 (71%)	-
Nonsterile gloves	2 (2, 2); 3 (100%)	2 (0, 6); 6 (86%)	6 (6, 6); 3 (100%)
Gauze	15 (5, 20); 3 (100%)	5 (5, 15); 7 (100%)	5 (5, 5); 3 (100%)
Dresing	1 (1, 1); 3 (100%)	1 (0, 1); 6 (86%)	1 (1, 1); 3 (100%)
Syringe	1 (1, 2); 3 (100%)	1 (1, 1); 7 (100%)	1 (1, 1); 3 (100%)
Kidney dish	1 (0, 1); 2 (67%)	1 (1, 1); 7 (100%)	1 (1, 1); 3 (100%)
Dressing pack[1]	1 (0, 1); 2 (67%)	0 (0, 1); 2 (29%)	1 (1, 1); 3 (100%)
Tape	15 (0, 25); 2 (67%)	1 (0, 10); 5 (71%)	3 (3, 5); 3 (100%)
Blueys	1 (0, 1); 2 (67%)	0 (0, 1); 2 (29%)	0 (0, 1); 1 (33%)
Tissue	2 (0, 8); 2 (67%)	0 (0, 6); 2 (29%)	6 (6, 8); 3 (100%)
Paper	1 (0, 2); 2 (67%)	0 (0, 1); 2 (29%)	1 (1, 1); 3 (100%)
Steristrip	0 (0, 1); 1 (33%)	-	-
Gallipot	0 (0, 1); 1 (33%)	1 (0, 1); 5 (71%)	-
Needle	0 (0, 1); 1 (33%)	1 (0, 1); 5 (71%)	-
Surface wipe	0 (0, 1); 1 (33%)	1 (0, 1); 5 (71%)	-
Biopsiblade (punch)	-	1 (1, 1); 7 (100%)	-
Examination bed sheet	-	1 (0, 1); 5 (71%)	-
Cotton tip applicator	-	0 (0, 2); 1 (14%)	2 (2, 2); 3 (100%)
Biopsiblade (shave)	-	-	1 (1, 1); 3 (100%)

[1] Contains disposable antiseptic field, tray, forceps, and gauze.

Supplementary Table 11. Materials used in the subgroup analysis exploring the excision of one lesion removed by either an excision, punch or shave biopsy.

Functional unit subgroup	Single-use materials	Reusable materials
One excision	Skin prep swab Drape Scalpel blade Specimen jar Specimen bag Suture Surgical mask Sterile gloves Nonsterile gloves Gauze Dressing Syringe Kidney dish Dressing pack[1] Tape Blueys Tissue Paper	Needle holder Scissors Forceps Scalpel handle
One punch biopsy	Skin prep swab Specimen jar Specimen bag Suture Surgical mask Sterile gloves Nonsterile gloves Gauze Dressing Syringe Kidney dish Tape Gallipot Needle Surface wipes Biopsiblade (punch) Examination bed sheet	Needle holder Scissors Forceps Tray
One shave biopsy	Skin prep swab Specimen jar Specimen bag Nonsterile gloves Gauze Dressing Syringe Kidney dish Dressing pack[1] Tape	-

	Tissue
	Paper
	Cotton tip applicator
	Biopsiblade (shave)

[1] Contains disposable antiseptic field, tray, forceps, and gauze.

Supplementary Table 12. Reusable and disposable materials used during an appointment for the excision of one or more than one lesions suspicious for melanoma or other skin cancer. Materials shaded green were used in more than 50% of appointments and were included in the primary analysis. All materials were included in the Monte Caro simulation.

	1 lesion (n=13)	**>1 lesion (n=13)**
Clinic, n (%)		
SMDC	6 (46%)	2 (15%)
MD	7 (54%)	11 (85%)
Number of lesions removed, n (%)		
1	13 (100%)	-
2	-	8 (62%)
3	-	2 (15%)
4	-	2 (15%)
5	-	1 (7·7%)
Type of excision for each lesion, n (%)		
Excision	3 (23%)	14 (40%)
Shave	3 (23%)	4 (11%)
Punch	7 (54%)	7 (49%)
Appointment duration, mins		
Median (min, max)	20 (15, 30)	40 (30, 45)
Reusable equipment, median (min, max) number used per appointment; n (%) appointments where item was used		
Needle holder	1 (0, 1); 10 (77%)	1 (0, 1); 12 (92%)
Forceps	1 (0, 2); 10 (77%)	1 (0, 2); 12 (92%)
Scissor	2 (0, 2); 8 (62%)	2 (0, 2); 12 (92%)
Scalpel handle	0 (0, 1); 6 (46%)	1 (0, 1); 9 (69%)
Tray	0 (0, 1); 6 (46%)	0 (0, 1); 2 (15%)
Gillies skin hook	-	0 (0, 1); 1 (8%)
Single-use equipment, mean (SD) number per appointment; n (%) appointments where item was used		

Skin prep swab	1 (1, 1); 13 (100%)	1 (1, 1); 13 (100%)
Specimen jar	1 (1, 1); 13 (100%)	2 (2, 5); 13 (100%)
Gauze	5 (5, 20); 13 (100%)	20 (5, 30); 13 (100%)
Syringe	1 (1, 2); 13 (100%)	3 (1, 10); 13 (100%)
Specimen bag	2 (0, 2); 12 (92%)	2 (1, 2); 13 (100%)
Nonsterile gloves	2 (0, 6) 12 (92%)	2 (2, 8); 13 (100%)
Kidney dish	1 (0, 1); 12 (92%)	1 (0, 1); 12 (92%)
Dresing	1 (0, 1); 12 (92%)	1 (0, 5); 9 (69%)
Suture	1 (0, 2); 10 (77%)	2 (0, 4); 12 (92%)
Tape	3 (0, 25); 10 (77%)	25 (1, 50); 13 (100%)
Surgical mask	1 (0, 2); 9 (69%)	2 (0, 2); 11 (85%)
Sterile gloves	2 (0, 2); 8 (62%)	2 (0, 4); 10 (77%)
Dressing pack[1]	1 (0, 1); 7 (54%)	1 (0, 1); 11 (85%)
Biopsiblade (punch)	1 (0, 1); 7 (54%)	1 (0, 2); 8 (62%)
Tissue	2 (0, 8); 7 (54%)	8 (0, 8); 10 (77%)
Paper	1 (0, 2); 7 (54%)	1 (0, 2); 11 (85%)
Drape	0 (0, 1); 6 (46%)	1 (0, 2); 11 (85%)
Gallipot	0 (0, 1); 6 (46%)	0 (0, 1); 2 (15%)
Needle	0 (0, 1); 6 (46%)	0 (0, 2); 3 (23%)
Surface wipe	0 (0, 1); 6 (46%)	0 (0, 1); 2 (15%)
Examination bed sheet	0 (0, 1); 5 (38%)	0 (0, 1); 1 (8%)
Blueys	0 (0, 1); 5 (38%)	1 (0, 2); 10 (77%)
Cotton tip applicator	0 (0, 2); 4 (31%)	0 (0, 4); 3 (23%)
Biopsiblade (shave)	0 (0, 1); 3 (23%)	0 (0, 1); 3 (23%)
Scalpel blade	0 (0, 1); 3 (23%)	1 (0, 1); 7 (54%)
Steristrip	0 (0, 1); 1 (8%)	0 (0, 1); 3 (23%)
Razor	-	0 (0, 1); 5 (38%)
Plastic bag	-	0 (0, 2); 2 (15%)
Curette	-	0 (0, 1); 1 (8%)

[1] Contains disposable antiseptic field, tray, forceps, and gauze.

Supplementary Table 13. Materials used in the subgroup analysis exploring the excision of one lesion or more than one lesion removed concurrently.

Functional unit subgroup	Single-use materials	Reusable materials
One lesion	Skin prep swab Specimen jar Gauze Syringe Specimen bag Nonsterile gloves Kidney dish Dressing Suture Tape Surgical mask Sterile gloves Dressing pack Biopsiblade (punch) Tissue Paper	Needle holder Forceps Scissors
More than one lesion	Skin prep swab Specimen jar Gauze Syringe Specimen bag Nonsterile gloves Kidney dish Dressing Suture Tape Surgical mask Sterile gloves Dressing pack Biopsiblade (punch) Tissue Paper Drape Blueys Scalpel blade	Needle holder Forceps Scissors Scalpel handle

Supplementary Table 14. Greenhouse gas emissions associated with one excision for a lesion suspicious for melanoma by the type of excision, and of the number of excisions. Table presents results of the subgroup analysis only.

	Type of excision (x1)			No. excisions	
	Excision	**Punch**	**Shave**	**1**	**>1**
Patient return transport, kg CO_2-e	10·72	10·72	10·72	10·72	10·72
Single-use instruments, kg CO_2-e	1·25	0·97	0·64	0·82	1·48
Reusable instruments, kg CO_2-e	0·23	1.97	-	0·19	0·23
Clinic room electricity use, kg CO_2-e	-	-	-	-	-
Total, kg CO_2-e	12.20	13.65	11.36	11·72	12.43

Supplementary Table 15. Table presents results of the Monte Carlo simulations (n=1000 runs, seed=360649). Mean, median, and 95% CI of greenhouse gas emissions each result reported in this study.

	Total GHG emissions, kg CO_2-e		
	Mean	**Median**	**95% CI**
Functional unit			
Full skin examination	10.23	4.53	1.93, 55.53
Total body photography	10.14	4.43	1.60, 54.54
Excision	12.33	6.08	2.92, 61.58
Type of excision, for the excision of one lesion only			
Excision biopsy	12.20	5.98	2.84, 61.45
Punch biopsy	13.15	6.46	3.41, 71.62
Shave biopsy	11.83	5.26	2.19, 63.84
Number of excisions			
One	11.67	5.32	2.41, 59.46
More than one	12.82	6.08	3.10, 58.47